

PUDASJARVI, Finland

WMO: 028670

Lat: 65.367N

Lon: 27.017E

Elev: 384

StdP: 14.49

Time Zone: 2.00 (EUE)

Period: 90-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-14.2	-23.6	1.5	-18.5	-18.8	2.0	-14.1	22.5	27.3	18.9	31.2	1.2	160	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	75.7	61.6	72.5	60.5	69.3	58.8	63.6	71.2	61.9	69.3	60.4	67.4	6.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	80.6	66.0	59.1	75.8	64.7	57.4	71.3	62.5	29.2	70.5	27.9	69.9	26.8	67.5	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.9	13.4	DB	-21.7	80.2	3.7	4.0	-24.4	83.1	-26.6	85.4	-28.7	87.6	-31.4	90.5	(4)
(5)			WB	-22.4	66.9	3.6	2.6	-25.0	68.8	-27.1	70.3	-29.2	71.8	-31.8	73.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	13.1	13.5	22.8	31.5	42.7	55.0	58.8	55.1	44.7	33.7	22.3	18.2		
	DBStd	18.66	13.87	14.24	9.09	8.91	7.13	6.54	5.25	5.76	6.53	9.35	10.38	13.18		
	HDD50	6366	1144	1022	843	555	250	25	2	17	185	504	832	987		
	HDD65	11182	1609	1442	1308	1004	691	306	200	310	609	969	1282	1452		
	CDD50	675	0	0	0	0	24	175	276	175	25	0	0	0		
	CDD65	18	0	0	0	0	0	6	9	3	0	0	0	0		
	CDH74	170	0	0	0	0	5	84	60	21	0	0	0	0		
	CDH80	15	0	0	0	0	0	7	8	0	0	0	0	0		
	WSAvg	5.9	5.3	5.5	6.5	6.1	6.4	6.2	5.8	5.3	5.8	6.3	5.6	5.7		
	PrecAvg	25.5	1.8	1.5	1.4	1.2	2.1	2.6	3.3	2.8	2.3	2.2	2.2	2.0		
	PrecMax	34.4	3.5	2.8	2.2	2.3	3.6	4.9	5.5	5.6	4.5	4.4	3.8	3.6		
(15) Precipitation	PrecMin	19.8	0.6	0.3	0.5	0.4	0.4	0.9	1.1	0.9	0.2	0.4	0.6	0.6		
	PrecStd	3.6	0.7	0.7	0.5	0.6	0.7	1.0	1.1	1.2	1.1	1.0	0.9	0.8		
	DB	38.2	39.5	41.0	58.5	73.1	80.7	79.6	76.7	65.6	52.4	40.9	38.4			
	MCWB	35.1	36.1	36.2	47.1	56.3	62.1	67.3	63.7	58.9	49.9	39.5	36.0			
	DB	34.6	36.7	38.4	52.1	66.1	76.4	75.0	72.9	60.9	49.7	38.5	36.1			
(23) Mean Coincident Wet Bulb Temperatures	2% MCWB	32.8	34.7	34.5	43.3	54.4	59.9	62.6	61.4	54.3	47.7	37.4	34.7			
	DB	32.6	34.3	36.1	48.0	60.7	72.1	71.9	69.5	57.5	47.5	36.9	34.8			
	5% MCWB	31.6	33.1	33.3	41.0	50.5	58.8	61.0	60.0	53.7	45.9	35.7	33.7			
	DB	30.3	32.6	34.1	44.0	55.6	67.9	68.9	66.2	54.5	45.3	34.9	33.3			
	10% MCWB	29.3	31.4	32.0	38.7	46.7	57.0	59.4	59.0	51.3	43.9	34.0	32.3			
	WB	35.2	37.0	37.6	48.0	59.0	64.6	67.8	65.2	59.3	50.5	40.4	36.9			
	MCDB	37.2	38.2	40.4	57.2	65.7	74.2	79.0	72.2	64.0	51.2	40.8	37.8			
	2% WB	33.3	35.0	35.1	44.1	54.5	61.9	64.0	62.9	56.7	48.2	38.0	35.1			
	5% MCDB	34.4	36.3	37.1	51.2	64.9	71.7	71.2	70.1	59.7	49.4	38.7	35.7			
	WB	31.8	33.3	33.8	41.5	50.7	60.2	62.1	61.2	53.9	46.0	36.1	33.9			
(31) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5% MCDB	32.6	34.3	35.8	46.7	59.2	70.6	68.9	67.9	56.6	47.2	36.9	34.7			
	WB	29.4	31.5	32.4	39.2	47.7	58.0	60.9	59.5	51.6	43.8	33.9	32.6			
	10% MCDB	30.3	32.7	34.1	43.8	54.2	66.5	67.2	65.5	54.1	45.2	34.9	33.5			
	MDBR	12.9	12.1	14.1	15.0	15.1	15.4	14.2	14.3	12.0	9.0	9.5	11.0			
	5% DB	10.4	8.4	13.4	18.6	24.7	23.2	20.3	20.4	14.7	10.4	6.5	7.9			
(36) Mean Daily Temperature Range	MCWBR	10.1	8.1	10.9	11.9	13.5	10.1	10.0	10.8	9.7	8.9	6.6	7.9			
	MCDBR	9.3	8.0	10.4	17.4	22.5	18.9	16.5	17.6	12.2	10.2	6.7	8.1			
	MCWBR	9.2	7.8	8.9	11.3	12.6	10.2	8.9	10.4	8.7	9.4	7.0	8.1			
	taub	0.201	0.272	0.302	0.338	0.343	0.344	0.364	0.345	0.321	0.297	0.347	N/A			
	taud	1.938	2.109	2.163	2.253	2.454	2.472	2.435	2.471	2.524	2.369	2.619	N/A			
(42) Clear-Sky Solar Irradiance	Ebn at Noon	89	189	241	262	274	276	266	260	242	186	85	N/A			
	Edh at Noon	7	18	28	33	30	30	30	27	20	15	6	N/A			
	RadAvg	34	190	640	1176	1495	1649	1553	1141	619	219	49	8			
(45) All-Sky Solar Radiation	RadStd	3	36	63	107	146	129	155	110	59	36	9	1			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	+1.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page